

Brainstorming & Idea Prioritization Template

Date	04 July 2026
Team ID	SOLO-CCFD-2026
Project Name	Credit Card Fraud Detection using Machine Learning
Maximum Marks	3 Marks

Step 1: Brainstorm and Idea Listing — Each team member lists out as many ideas as possible without judging them at this stage.

S.No	Team Member	Idea / Suggestion	Category	Group No.
1	Ayusha Sanjuktha	Use PCA-transformed transaction features with a supervised classifier to flag fraud in real time	Modeling	1
2	Ayusha Sanjuktha	Handle extreme class imbalance (0.17% fraud) using SMOTE / class-weighting before training	Data Prep	1
3	Ayusha Sanjuktha	Compare Logistic Regression, Decision Tree, Random Forest and XGBoost, pick best by recall & AUPRC	Modeling	1
4	Ayusha Sanjuktha	Build a Flask web app where an analyst enters transaction values and gets an instant fraud/genuine prediction	Application	2
5	Ayusha Sanjuktha	Deploy the trained model via IBM Watson Machine Learning for scalable, cloud-hosted scoring	Deployment	2
6	Ayusha Sanjuktha	Add a batch-upload mode so a compliance officer can screen many transactions in one CSV at once	Application	2

Step 2: Idea Prioritization — Rate each grouped idea on feasibility and importance, then select the final idea(s) to move forward with.

Group No.	Final Idea	Feasibility	Importance	Priority Selected
1	Real-time single-transaction fraud scoring using an ensemble of 4 ML models trained on PCA features with SMOTE-balanced data	High	High	Yes
2	Flask + IBM Watson ML deployment for real-time and batch fraud screening with an analyst-facing UI	High	High	Yes